Learning from an operational utility-scale Li-Polymer battery system in a PV plant

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Outline

- Research at UQ Solar
- About UQ Gatton Solar Research Facility (GSRF)
  - PV Arrays
  - Utility-scale battery storage system
- Battery operation analysis in:
  - Different operation modes
  - Ramp-rate control mode
- Conclusions

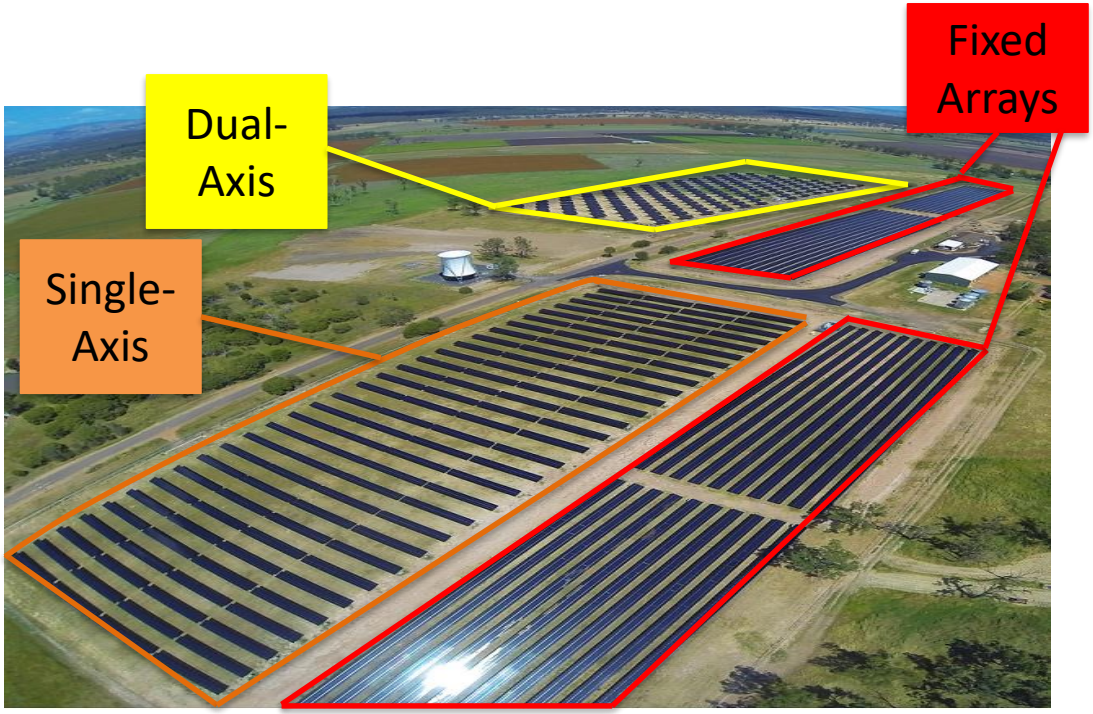
Research at UQ Solar

- People
  - Prof. Tapan Saha, Director
  - 1 ARC DECRA Fellow
  - 3 Adv. QLD Fellows
  - 3 Research Fellows
  - 5 PhD students
  - 1 MPhil student

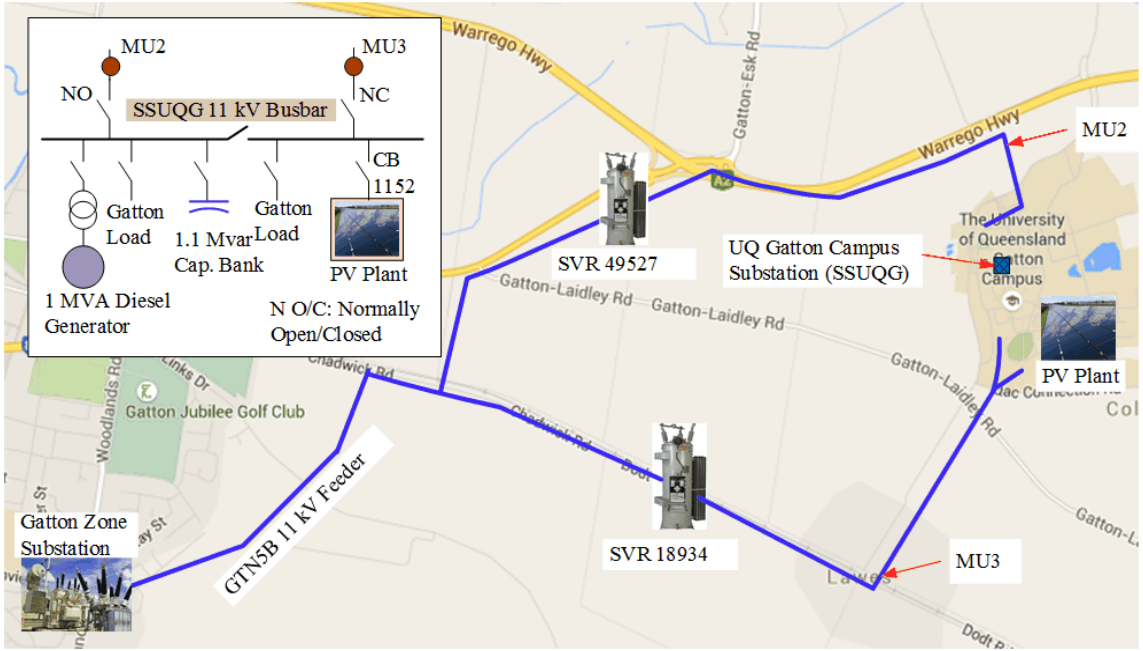
- Research Themes
  - Economics
    - Solar PV yield assessment
    - Assessing commercial viability of PV in the NEM
  - Power system operation and control
    - PV voltage control strategies
    - PV and battery integration in remote communities
    - Peak shaving and demand response
  - Condition monitoring
    - Battery/PV degradation modelling
    - Monitoring plant performance

About GSRF PV Plant

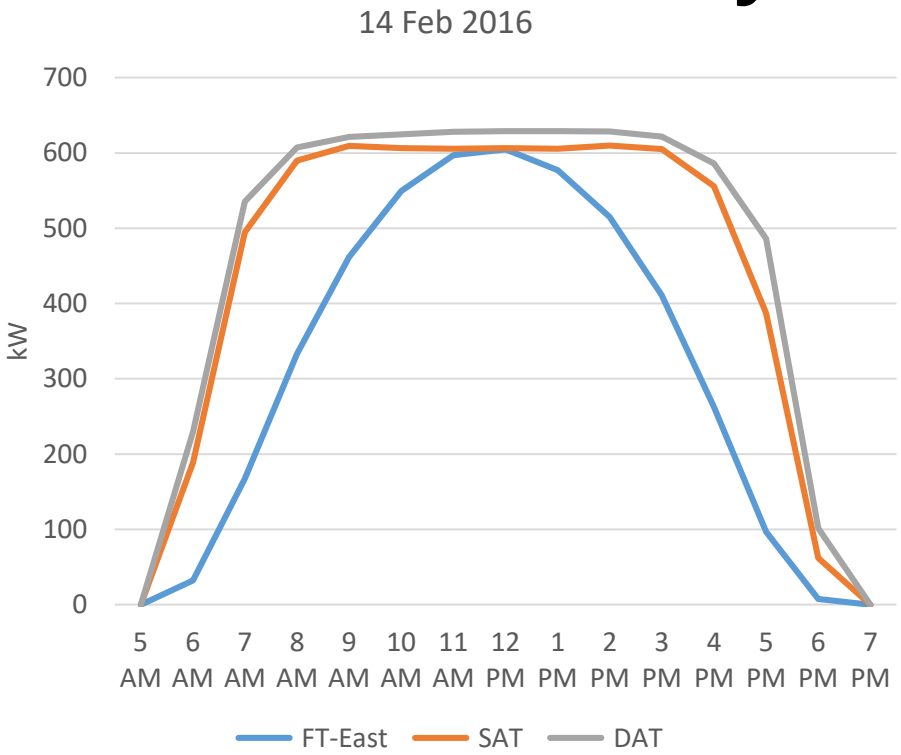
- Commissioned in March 2015
- 3.275 MWp capacity
- 37,000 Cd-Te PV panels
- 5 × centralized SMA inverters 720 kWp capped at 630 kWp
- Fixed, single, and dual axis tracking systems

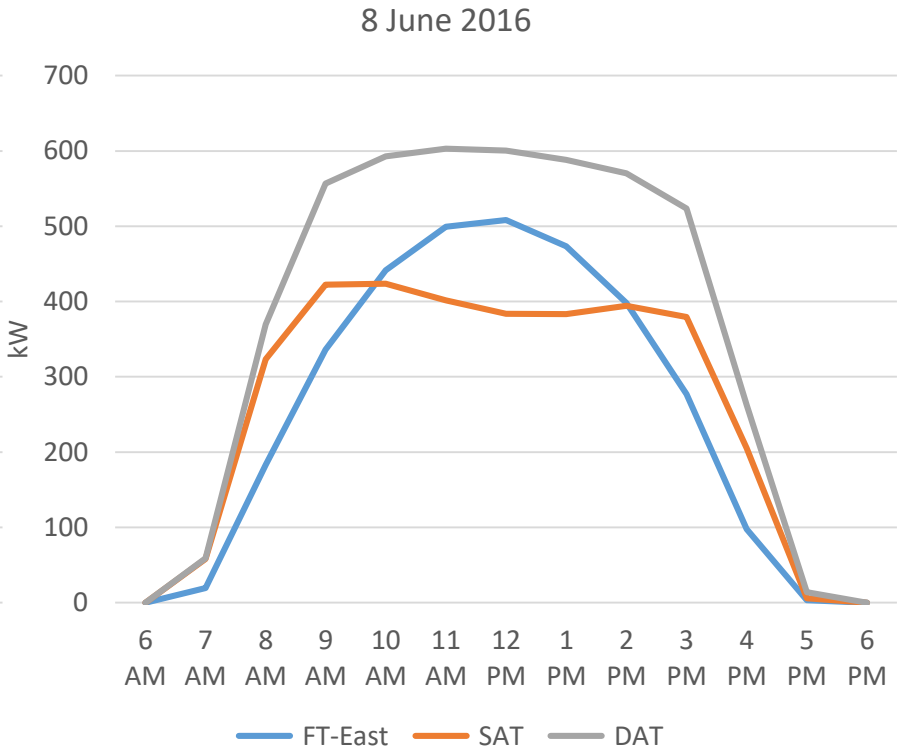


Connection to the Grid



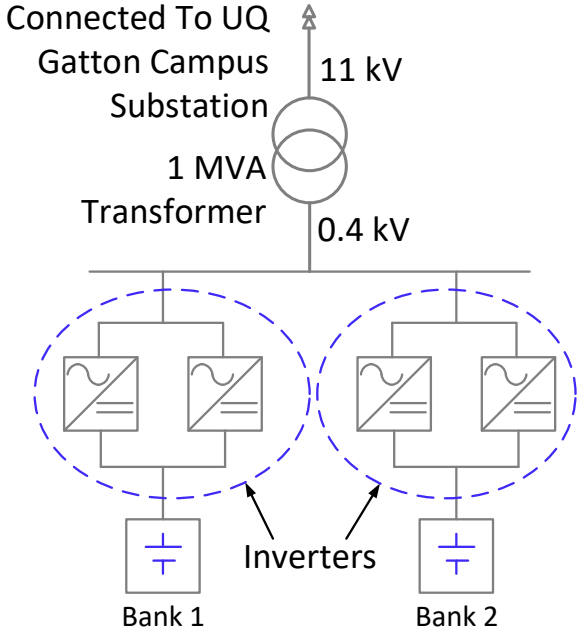
PV Arrays Performance





About GSRF Battery Storage

- 600 kW, 760 kWh Lithium-Polymer BESS (75Ah per cell, 4000 cycles at 80% DOD)
- 576~748 V DC
- Interfaced by 4×300 kVA VACON Inverters with 415 V, 3-phase AC output
- Capable to source/sink reactive power up to 0.9 PF
- Operates in 10 pre-defined modes



Battery Performance

Operation mode	% of time	Total charge MWh (cycles)	Total discharge MWh (cycles)	Average Abs. power	Std. Abs. Power
Idle	75.6	--	--	--	--
Charge	14.5	96.2 (158.2)	--	39.7	39.1
Discharge	7.8	--	68.5 (112.7)	65.8	24.6
Charge/Discharge	2.2	1.1 (1.8)	11.0 (18.1)	48.2	69.76

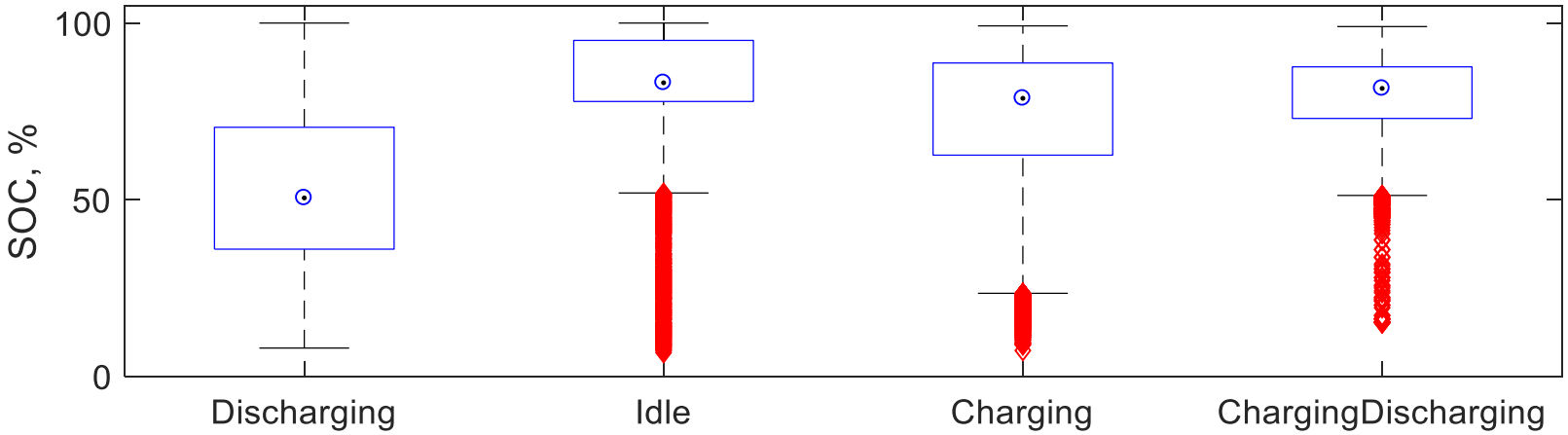
Ramp-rate control

Total # of cycles ≈ 291

RT efficiency + self-discharge ≈ 82%

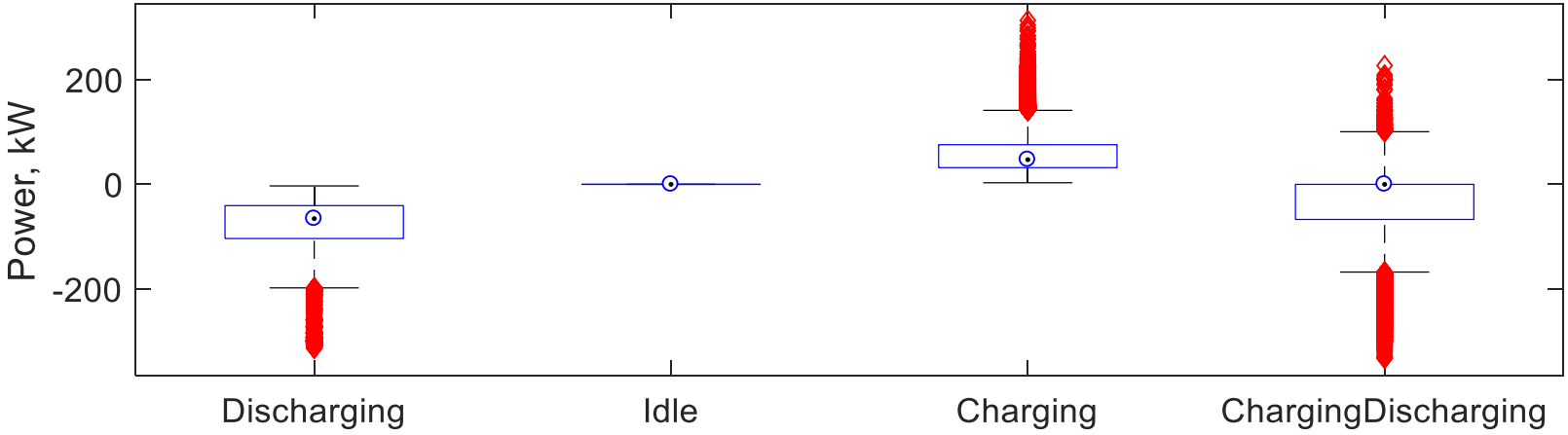
Battery Performance

State-of-charge of the battery in each operation mode:



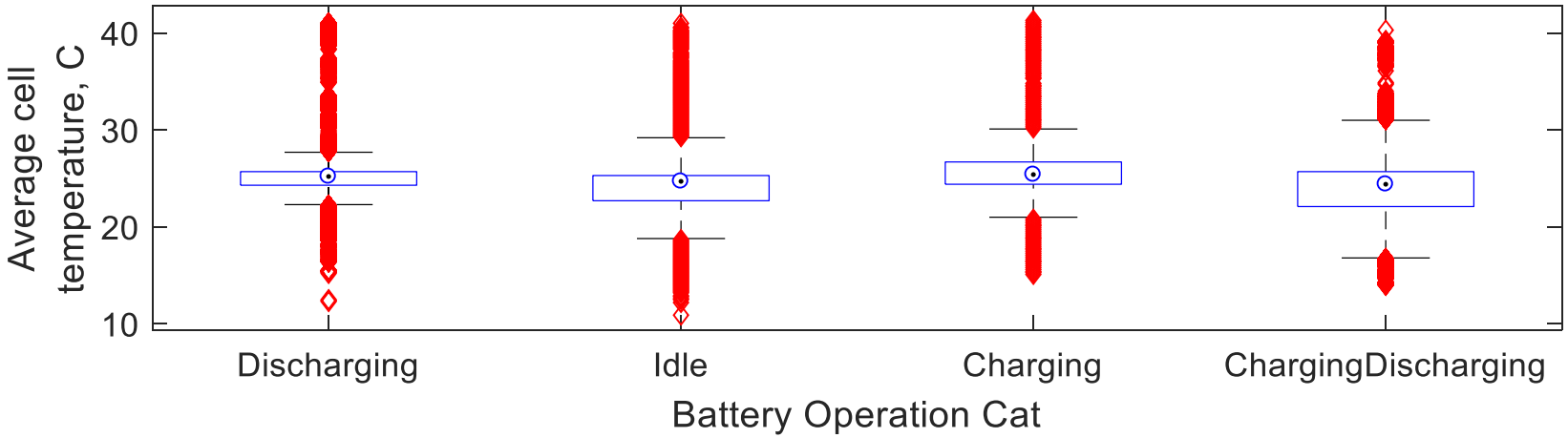
Battery Performance

Power magnitude of the battery in each operation mode:



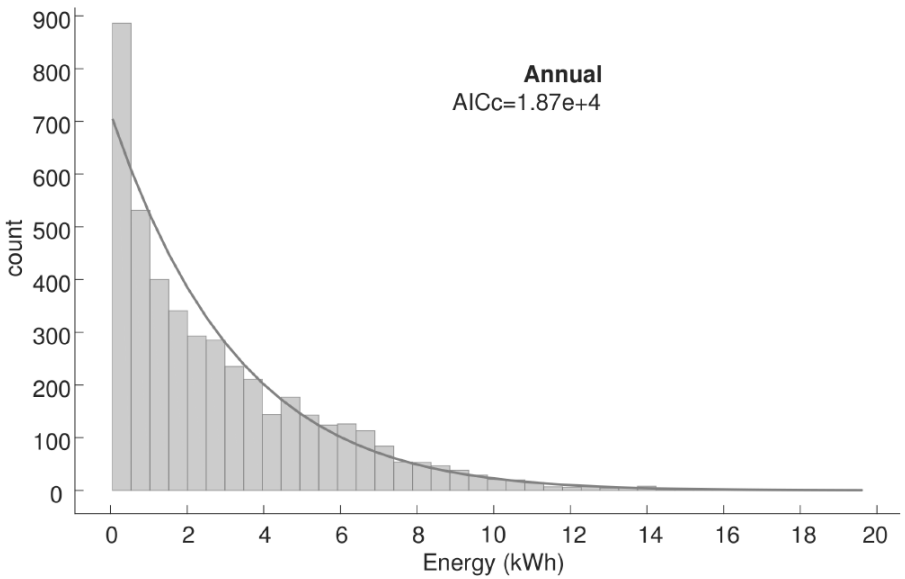
Battery Performance

Average battery cell temperature for each operation mode:



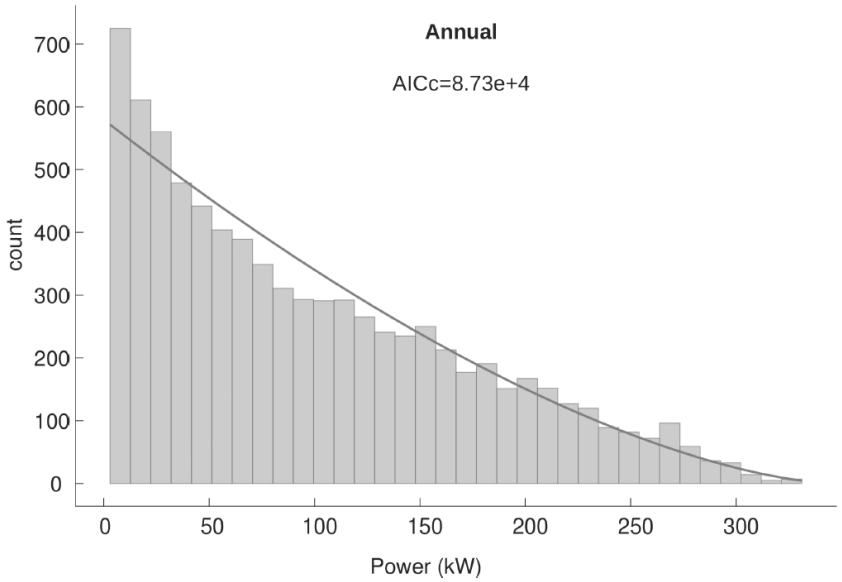
Battery in Ramp Control Mode

- 7,928 minutes of operation (more than 11 days accumulatively)
- 21.3 cycles in total:
  - About 1.8 cycles per aggregated days, which is more than about half a cycle per day for battery operation
- Energy of 50% of the events smaller than 2.1 kWh → case for super-capacitor



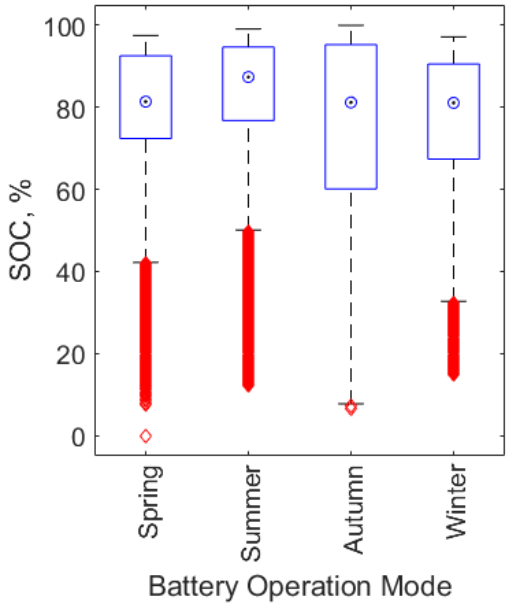
Battery in Ramp Control Mode

- 86% of power samples are less than 150 kW
- Strong correlation between energy and power samples
- The two observations further justifies application of hybrid storage system

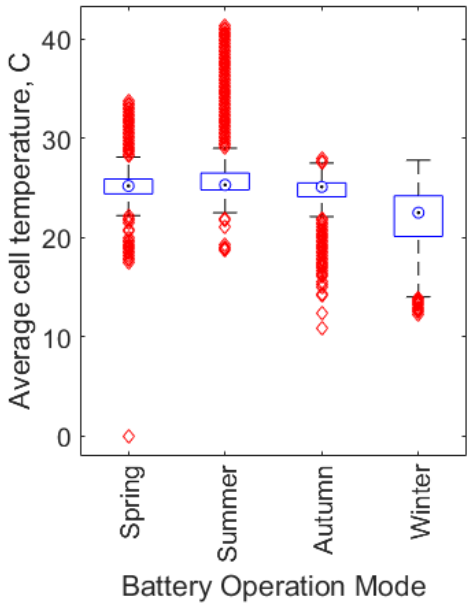


Battery Seasonal Performance

Battery SOC in different seasons



Battery cell temperature in different seasons



Conclusions

- Stacked application of the battery should be considered for better rate-of-return on investment
- Battery capable of high DoD are needed for smoothing PV generation (or the battery should be oversized significantly)
- Seasonal changes in the battery operation management system would be beneficial to
  - Improve SOC of the battery,
  - Provide better reliability of the plant operation, and
  - Avoid battery unnecessary degradation
- Battery cell temperature shows larger deviation during ramp-rate control mode
  - It can be used to design appropriate cooling systems

Acknowledgment

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Thanks for your attention

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